Geochemical signatures in the long-lived marine bivalve Arctica islandica show intense warming in the down east coastal region of the Gulf of Maine

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The Gulf of Maine (GoM) is undergoing rapid environmental changes, and this system is projected to become increasingly stressed in the coming decades. In this study, we used the growth and geochemical signatures from the long-lived marine bivalve Arctica islandica collected in 70 to 80 m water depth from the Down East coastal region in the GoM (Jonesport, ME, USA) to evaluate past climatic and hydrographic variability. The master shell growth chronology is precisely dated via crossdating methods and spans from 1953 to 2023 CE with shell growth variability being highly coherent among the population, indicating that environmental conditions are driving growth. The Jonesport stable oxygen isotope ($\delta^{18}O_{shell}$) series spans from 1956-2020 CE and demonstrates a secular decrease in $\delta^{18}O_{shell}$ values over time, indicating an increase in seawater temperature and/or freshening of waters over the past ~75 years. Radiocarbon dated shells and $\delta^{18}O_{shell}$ data from the 17th century provide a pre-industrial “floating” baseline context for the changes noted during the last 75 years. To further constrain the relative influences temperature and salinity (source water oxygen isotope values calculated) variations on the $\delta^{18}O_{shell}$ record since ~ 1970, output from the VIKING20X model (1/20° Atlantic configuration nested in 1/4° global ocean, forced by the 55-year Japanese Atmospheric Reanalysis product) is used. Although broadly consistent with other temperature records in the region, the $\delta^{18}O_{shell}$ series indicates that the warming since 1956 in the Down East coastal region of the GoM is large (> 3 °C) and will likely have negative ecological impacts.